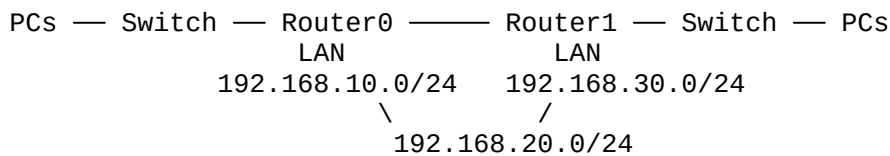


Absolutely. Since we now know your **actual Packet Tracer interface names**, here is the complete setup from beginning to end for your two-router EIGRP + DHCP topology.

I'll separate **physical actions** from **copy-paste CLI commands**, so you know exactly when to add the module, connect cables, configure routers, and configure PCs.

Complete Packet Tracer Setup

Your final topology should conceptually be:



Your actual serial interfaces are:

Device	Interface	Network	IP
Router0	Fa0/0	192.168.10.0/24	192.168.10.1
Router0	Serial0/3/0	192.168.20.0/24	192.168.20.1
Router1	Serial0/1/0	192.168.20.0/24	192.168.20.2
Router1	Fa0/0	192.168.30.0/24	192.168.30.1

STEP 1 — Place the devices

Add:

- 2 × **Cisco 2811 routers**
- 2 × **switches**
- Your required PCs

Your lab starts with Cisco 2811 routers, switches, and PCs.

STEP 2 — Add the serial modules

Do this before configuring the serial interfaces.

Router0

1. Click **Router0**

2. Go to **Physical**
3. Turn the router **OFF**
4. Remove the Ethernet module you previously installed.
5. Select:

WIC-1T

6. Drag WIC-1T into an available WIC slot.
7. Turn Router0 **ON**.

Router1

Do the exact same thing:

1. Physical
2. Power OFF
3. Add **WIC-1T**
4. Power ON

Your lab handout mentions adding a WIC module before configuration, although the specific WIC-1ENET listed there is an Ethernet card and does not provide the serial interface required by the later commands.

For **your Packet Tracer setup**, WIC-1T is what gave you the required serial interface.

STEP 3 — Connect the cables

Connect:

Router0 → Router1

Use a **Serial DCE cable**:

```
Router0 Serial0/3/0
      |
      | Serial
      |
Router1 Serial0/1/0
```

Then:

```
Router0 Fa0/0 — Switch0 — PCs
Router1 Fa0/0 — Switch1 — PCs
```

Use the appropriate Ethernet cable for the router-to-switch and PC-to-switch connections.

STEP 4 — Configure Router0

Go to:

Router0 → CLI

Paste this entire block:

```
enable
configure terminal

interface fastEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface serial 0/3/0
ip address 192.168.20.1 255.255.255.0
clock rate 128000
no shutdown
exit

ip dhcp excluded-address 192.168.10.1

ip dhcp pool LAN10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
exit

router eigrp 10
network 192.168.10.0 255.255.255.0
network 192.168.20.0 255.255.255.0
exit

exit
copy running-config startup-config
```

When you see:

Destination filename [startup-config]?

press **Enter**.

What Router0 is doing

Router0 now has:

```
Fa0/0 → 192.168.10.1
Serial0/3/0 → 192.168.20.1
```

It also acts as a **DHCP server** for the 192.168.10.0/24 LAN.

And EIGRP tells Router0:

"Advertise my 192.168.10.0 and 192.168.20.0 networks to my EIGRP neighbor."

The EIGRP configuration follows your lab's Router0 configuration.

STEP 5 — Check Router0

Run:

```
show ip interface brief
```

You want to see approximately:

FastEthernet0/0	192.168.10.1	up	up
Serial0/3/0	192.168.20.1	up	up

If Serial says:

down down

don't continue blindly. The serial cable/DCE side needs to be checked.

STEP 6 — Configure Router1

Go to:

Router1 → CLI

Paste:

```
enable
configure terminal
```

```
interface fastEthernet 0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

```
interface serial 0/1/0
ip address 192.168.20.2 255.255.255.0
no shutdown
exit
```

```
ip dhcp excluded-address 192.168.30.1
```

```
ip dhcp pool LAN30
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
exit
```

```
router eigrp 10
```

```
network 192.168.20.0 255.255.255.0
network 192.168.30.0 255.255.255.0
exit
```

```
exit
copy running-config startup-config
```

Press **Enter** when asked for the destination filename.

The lab's intended Router1 addressing is 192.168.30.1 on the LAN and 192.168.20.2 on the serial link, with those two networks advertised through EIGRP.

STEP 7 — Check Router1

Run:

```
show ip interface brief
```

You want:

FastEthernet0/0	192.168.30.1	up	up
Serial0/1/0	192.168.20.2	up	up

STEP 8 — Configure the PCs using DHCP

Do NOT manually type IP addresses into the PCs.

For every PC:

PC → Desktop → IP Configuration → DHCP

Wait a few seconds.

PCs connected to Router0

They should receive something like:

```
IP address:      192.168.10.x
Subnet mask:     255.255.255.0
Default gateway: 192.168.10.1
```

PCs connected to Router1

They should receive:

```
IP address:      192.168.30.x
Subnet mask:     255.255.255.0
Default gateway: 192.168.30.1
```

The DHCP configuration is something we added to automate the PC addressing; the original lab itself focuses on EIGRP and manually specified router interface addressing.

STEP 9 — Verify DHCP

On **Router0**:

```
show ip dhcp binding
```

You should see the PCs that received 192.168.10.x addresses.

On **Router1**:

```
show ip dhcp binding
```

You should see the PCs that received 192.168.30.x addresses.

STEP 10 — Check the EIGRP neighbor

This is VERY important.

On **Router0**:

```
show ip eigrp neighbors
```

You want to see Router1 listed as a neighbor.

Then on Router1:

```
show ip eigrp neighbors
```

You should see Router0.

If there is **no neighbor**, the first thing to check is the serial link.

STEP 11 — Check the routing table

On Router0:

```
show ip route
```

You should eventually see something like:

```
C    192.168.10.0/24
C    192.168.20.0/24
D    192.168.30.0/24
```

The important one is:

```
D    192.168.30.0/24
```

D means the route was learned through **EIGRP**. Your lab uses `show ip route` for this verification and shows `D 192.168.30.0/24` as the EIGRP-learned network.

On Router1:

```
show ip route
```

You should see:

```
C    192.168.30.0/24
C    192.168.20.0/24
D    192.168.10.0/24
```

STEP 12 — Test connectivity

Now test in this order.

Test 1 — Router0 → Router1

From Router0:

```
ping 192.168.20.2
```

You want successful replies.

Test 2 — Router1 → Router0

From Router1:

```
ping 192.168.20.1
```

Test 3 — Router0 → Router1's LAN

From Router0:

```
ping 192.168.30.1
```

Test 4 — Router1 → Router0's LAN

From Router1:

```
ping 192.168.10.1
```

Test 5 — PC-to-PC

From a PC on the 192.168.10.x side, ping a PC on the 192.168.30.x side:

ping 192.168.30.x

If this works, you have demonstrated the whole setup:

```
DHCP
↓
PC receives IP
↓
Default Gateway
↓
Router
↓
EIGRP
↓
Other Router
↓
Remote LAN
↓
Remote PC
```

The whole sequence to remember

For your lab, the order is:

1. Add routers/switches/PCs
↓
2. Power OFF routers
↓
3. Add WIC-1T modules
↓
4. Power ON routers
↓
5. Connect serial + Ethernet cables
↓
6. Configure Router0 interfaces
↓
7. Configure Router1 interfaces
↓
8. Configure DHCP on both routers
↓
9. Configure EIGRP on both routers
↓
10. Set all PCs to DHCP
↓
11. Check serial interfaces
↓
12. Check EIGRP neighbors
↓
13. Check routing tables
↓
14. Ping across the network
↓
15. Save configurations

One important thing for your current Packet Tracer setup

You **already installed the WIC-1T modules and configured the IPs**, so **do not redo Steps 1–7**. You're currently around Step 8–10.

Your next useful checks are:

Router0:

```
show ip interface brief
show ip eigrp neighbors
show ip route
```

Router1:

```
show ip interface brief
show ip eigrp neighbors
show ip route
```

If you paste those outputs, I can tell you exactly whether your topology is fully working or where it is failing.

Yes. Since your previous setup uses **2811 routers + WIC-1T**, I'll keep the same style and use **DHCP for PCs + EIGRP for routing**.

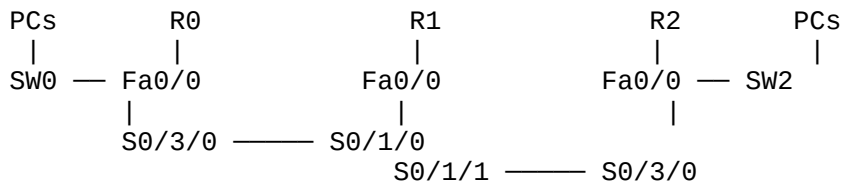
I'll give you two complete topologies:

1. **3 routers in a line:** R0 — R1 — R2
2. **3 routers connected to each other:** triangle R0 — R1 — R2 — R0

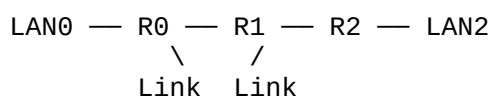
For both, **router interfaces use static IPs**, while PCs get IPs through DHCP.

1. Three routers connected in a LINE

Topology:



Conceptually:



We'll use:

Router	Interface	IP
R0	Fa0/0	192.168.10.1
R0	Serial	192.168.20.1
R1	Serial to R0	192.168.20.2
R1	Fa0/0	192.168.30.1
R1	Serial to R2	192.168.40.1
R2	Serial to R1	192.168.40.2
R2	Fa0/0	192.168.50.1

Important: Your actual serial interface numbers depend on where you insert the WIC-1T. Use `show ip interface brief` to see them. Below I use example names. Replace them with your actual names if they differ.

STEP 1 — Add modules

For **all 3 routers**:

1. Open router → **Physical**
2. Power **OFF**
3. Add **WIC-1T**
4. Power **ON**

R1 needs **two serial interfaces**, because it connects to both R0 and R2.

So for R1, you'll need **two WIC-1T modules** if your router configuration only provides one serial interface per module.

Check:

```
show ip interface brief
```

R1 needs to have two serial interfaces.

STEP 2 — Configure Router0

Assume R0 serial interface is Serial0/3/0.

```
enable
configure terminal
```

```
interface fastEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit
```

```
interface serial 0/3/0
ip address 192.168.20.1 255.255.255.0
clock rate 128000
no shutdown
exit

ip dhcp excluded-address 192.168.10.1

ip dhcp pool LAN10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
exit

router eigrp 10
network 192.168.10.0 255.255.255.0
network 192.168.20.0 255.255.255.0
exit

exit
copy running-config startup-config
```

STEP 3 — Configure Router1

R1 has **three important interfaces**:

Serial → R0
FastEthernet → Switch/PCs
Serial → R2

Assume:

Serial0/1/0 → R0
FastEthernet0/0 → LAN
Serial0/1/1 → R2

Configure:

```
enable
configure terminal

interface fastEthernet 0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit

interface serial 0/1/0
ip address 192.168.20.2 255.255.255.0
no shutdown
exit

interface serial 0/1/1
ip address 192.168.40.1 255.255.255.0
clock rate 128000
no shutdown
exit
```

```
ip dhcp excluded-address 192.168.30.1
```

```
ip dhcp pool LAN30  
network 192.168.30.0 255.255.255.0  
default-router 192.168.30.1  
exit
```

```
router eigrp 10  
network 192.168.20.0 255.255.255.0  
network 192.168.30.0 255.255.255.0  
network 192.168.40.0 255.255.255.0  
exit
```

```
exit  
copy running-config startup-config
```

Important about clock rate

Only the **DCE side** of each serial link should have the clock rate.

So if R1 isn't the DCE side of the R1 → R2 cable, don't use:

```
clock rate 128000
```

You can check with:

```
show controllers serial 0/1/1
```

If it says **DCE**, use the clock rate.

STEP 4 — Configure Router2

Assume R2's serial interface is Serial0/3/0.

```
enable  
configure terminal
```

```
interface fastEthernet 0/0  
ip address 192.168.50.1 255.255.255.0  
no shutdown  
exit
```

```
interface serial 0/3/0  
ip address 192.168.40.2 255.255.255.0  
no shutdown  
exit
```

```
ip dhcp excluded-address 192.168.50.1
```

```
ip dhcp pool LAN50  
network 192.168.50.0 255.255.255.0  
default-router 192.168.50.1  
exit
```

```
router eigrp 10
network 192.168.40.0 255.255.255.0
network 192.168.50.0 255.255.255.0
exit
```

```
exit
copy running-config startup-config
```

STEP 5 — Configure PCs

Every PC:

PC → Desktop → IP Configuration → DHCP

R0-side PCs should get:

```
192.168.10.x
Gateway: 192.168.10.1
```

R1-side PCs:

```
192.168.30.x
Gateway: 192.168.30.1
```

R2-side PCs:

```
192.168.50.x
Gateway: 192.168.50.1
```

STEP 6 — Check EIGRP

On **R0**:

```
show ip eigrp neighbors
```

R0 should have **R1** as a neighbor.

On **R1**:

```
show ip eigrp neighbors
```

R1 should have **two neighbors**:

```
R0
R2
```

On **R2**:

```
show ip eigrp neighbors
```

R2 should have **R1**.

STEP 7 — Check routing

On R0:

show ip route

You should eventually see:

```
C    192.168.10.0/24
C    192.168.20.0/24
D    192.168.30.0/24
D    192.168.40.0/24
D    192.168.50.0/24
```

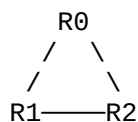
The D routes are learned through EIGRP.

On R2, you should see the opposite direction:

```
D    192.168.10.0/24
D    192.168.20.0/24
D    192.168.30.0/24
C    192.168.40.0/24
C    192.168.50.0/24
```

2. Three routers connected WITH EACH OTHER

This is the **triangle topology**.



Each router has a direct connection to the other **two routers**.

So there are **three separate router-to-router networks**.

We'll use:

```
R0 ↔ R1 = 192.168.20.0/24
R1 ↔ R2 = 192.168.40.0/24
R2 ↔ R0 = 192.168.60.0/24
```

LANs:

```
R0 LAN = 192.168.10.0/24
```

R1 LAN = 192.168.30.0/24
R2 LAN = 192.168.50.0/24

IP plan

Router	Connection	IP
R0	LAN	192.168.10.1
R0	R1 link	192.168.20.1
R0	R2 link	192.168.60.1
R1	LAN	192.168.30.1
R1	R0 link	192.168.20.2
R1	R2 link	192.168.40.1
R2	LAN	192.168.50.1
R2	R1 link	192.168.40.2
R2	R0 link	192.168.60.2

STEP 1 — Modules

Each router needs enough serial interfaces for **two router-to-router connections**.

R0 → 2 serial interfaces
R1 → 2 serial interfaces
R2 → 2 serial interfaces

So install enough **WIC-1T** modules to provide two serial interfaces on each router.

Then check each router:

```
show ip interface brief
```

You need two serial interfaces on each router.

STEP 2 — Router0

Assume:

Serial0/3/0 → R1
Serial0/3/1 → R2

Commands:

```
enable  
configure terminal
```

```
interface fastEthernet 0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
exit

interface serial 0/3/0
ip address 192.168.20.1 255.255.255.0
clock rate 128000
no shutdown
exit

interface serial 0/3/1
ip address 192.168.60.1 255.255.255.0
clock rate 128000
no shutdown
exit

ip dhcp excluded-address 192.168.10.1

ip dhcp pool LAN10
network 192.168.10.0 255.255.255.0
default-router 192.168.10.1
exit

router eigrp 10
network 192.168.10.0 255.255.255.0
network 192.168.20.0 255.255.255.0
network 192.168.60.0 255.255.255.0
exit

exit
copy running-config startup-config
```

Again, keep clock rate only on interfaces that are actually **DCE**.

STEP 3 — Router1

Assume:

```
Serial0/1/0 → R0
Serial0/1/1 → R2
```

```
enable
configure terminal
```

```
interface fastEthernet 0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

```
interface serial 0/1/0
ip address 192.168.20.2 255.255.255.0
no shutdown
exit
```

```
interface serial 0/1/1
ip address 192.168.40.1 255.255.255.0
clock rate 128000
no shutdown
exit

ip dhcp excluded-address 192.168.30.1

ip dhcp pool LAN30
network 192.168.30.0 255.255.255.0
default-router 192.168.30.1
exit

router eigrp 10
network 192.168.20.0 255.255.255.0
network 192.168.30.0 255.255.255.0
network 192.168.40.0 255.255.255.0
exit

exit
copy running-config startup-config
```

STEP 4 — Router2

Assume:

Serial0/3/0 → R1
Serial0/3/1 → R0

```
enable
configure terminal

interface fastEthernet 0/0
ip address 192.168.50.1 255.255.255.0
no shutdown
exit

interface serial 0/3/0
ip address 192.168.40.2 255.255.255.0
no shutdown
exit

interface serial 0/3/1
ip address 192.168.60.2 255.255.255.0
no shutdown
exit

ip dhcp excluded-address 192.168.50.1

ip dhcp pool LAN50
network 192.168.50.0 255.255.255.0
default-router 192.168.50.1
exit

router eigrp 10
```

```
network 192.168.40.0 255.255.255.0
network 192.168.50.0 255.255.255.0
network 192.168.60.0 255.255.255.0
exit
```

```
exit
copy running-config startup-config
```

STEP 5 — PCs

Set every PC to:

Desktop → IP Configuration → DHCP

You should get:

```
R0 PCs → 192.168.10.x
R1 PCs → 192.168.30.x
R2 PCs → 192.168.50.x
```

STEP 6 — Verify the triangle

On **each router**:

```
show ip eigrp neighbors
```

This time, each router should have **2 EIGRP neighbors**.

R0

```
R1
R2
```

R1

```
R0
R2
```

R2

```
R0
R1
```

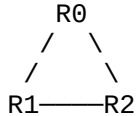
That's the major difference between the two topologies.

Line

```
R0 — R1 — R2
```

R0 → 1 neighbor
R1 → 2 neighbors
R2 → 1 neighbor

Triangle



R0 → 2 neighbors
R1 → 2 neighbors
R2 → 2 neighbors

Very important: don't blindly use my serial names

Because you've already seen that your Packet Tracer generated:

R0 → Serial0/3/0
R1 → Serial0/1/0

your **3-router serial interface numbers may be different** depending on the module slots.

Before pasting the serial portions, run on **each router**:

```
show ip interface brief
```

Then substitute the actual interface names.

The **IP addresses and EIGRP network commands remain exactly as shown above.**

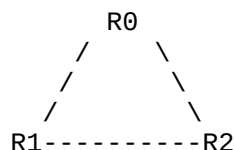
Also, the original lab's EIGRP convention is `router eigrp 10` with the relevant connected networks advertised using `network` commands.

Absolutely. We'll use the **3-router triangle topology**, with **DHCP for every PC**, and I'll give you **copy-paste commands in the exact order**.

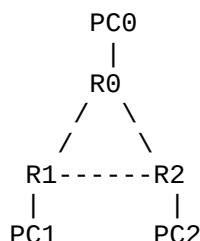
Your PDF's two experiments are RIP and OSPF; for OSPF it specifically mentions adding a **WIC-1ENET module** when more than two routers are used.

1. Topology

Build this:



And connect one PC to each router:



So there are **6 important links**:

- PC0 ↔ R0
- PC1 ↔ R1
- PC2 ↔ R2
- R0 ↔ R1
- R0 ↔ R2
- R1 ↔ R2

2. Addressing plan

We'll use this simple addressing scheme.

Router-to-router links

Connection	Network	First router	Second router
R0 ↔ R1	192.168.10.0/24	R0 = 192.168.10.1	R1 = 192.168.10.2
R0 ↔ R2	192.168.20.0/24	R0 = 192.168.20.1	R2 = 192.168.20.2

Connection	Network	First router	Second router
R1 ↔ R2	192.168.30.0/24	R1 = 192.168.30.1	R2 = 192.168.30.2

PC LANs

Router	LAN	Router's LAN IP
R0	192.168.1.0/24	192.168.1.1
R1	192.168.2.0/24	192.168.2.1
R2	192.168.3.0/24	192.168.3.1

The PCs will get their IPs **automatically through DHCP**.

3. Add the modules **FIRST**

Before entering commands, configure the physical router interfaces.

Use **Cisco 2901 routers**, as specified in your PDF.

For each router:

1. Click the router.
2. Go to **Physical**.
3. Turn the router **OFF**.
4. Find **WIC-1ENET**.
5. Drag the **WIC-1ENET** module into an available slot.
6. Turn the router **ON** again.

Your PDF specifically says to add WIC-1ENET when using more than two routers in the OSPF experiment.

Important

After inserting the module, your available interface names may depend on the Packet Tracer router/module configuration.

Check the interface names with:

```
show ip interface brief
```

If your router has GigabitEthernet0/0, GigabitEthernet0/1, and an additional Ethernet interface, use the actual name Packet Tracer shows.

4. Connect the cables

Use Ethernet connections.

For example:

```
R0 G0/0 → PC0/Switch
R0 G0/1 → R1
R0 additional interface → R2
```

```
R1 G0/0 → PC1/Switch
R1 G0/1 → R0
R1 additional interface → R2
```

```
R2 G0/0 → PC2/Switch
R2 G0/1 → R0
R2 additional interface → R1
```

The exact third-interface name can vary, so check `show ip interface brief` before using it.

PART A — RIP

We'll configure the triangle using **RIP version 2**, matching your Experiment 7. Your PDF uses `router RIP, version 2`, and `network` statements.

5. Configure Router 0

First:

```
enable
show ip interface brief
```

Identify your three interfaces.

Assuming they are:

```
GigabitEthernet0/0
GigabitEthernet0/1
FastEthernet0/0
```

we'll use:

```
G0/0 → PC0
G0/1 → R1
Fa0/0 → R2
```

Then enter:

```
enable
```

```
configure terminal
```

```
interface gigabitEthernet 0/0  
ip address 192.168.1.1 255.255.255.0  
no shutdown  
exit
```

```
interface gigabitEthernet 0/1  
ip address 192.168.10.1 255.255.255.0  
no shutdown  
exit
```

```
interface fastEthernet 0/0  
ip address 192.168.20.1 255.255.255.0  
no shutdown  
exit
```

Now DHCP for PC0:

```
ip dhcp excluded-address 192.168.1.1 192.168.1.10
```

```
ip dhcp pool LAN_R0  
network 192.168.1.0 255.255.255.0  
default-router 192.168.1.1  
exit
```

Now RIP:

```
router rip  
version 2  
no auto-summary  
network 192.168.1.0  
network 192.168.10.0  
network 192.168.20.0  
exit
```

```
end  
write
```

6. Configure Router 1

Assume:

```
G0/0 → PC1  
G0/1 → R0  
Fa0/0 → R2
```

Enter:

```
enable  
configure terminal
```

```
interface gigabitEthernet 0/0  
ip address 192.168.2.1 255.255.255.0
```

```
no shutdown
exit
```

```
interface gigabitEthernet 0/1
ip address 192.168.10.2 255.255.255.0
no shutdown
exit
```

```
interface fastEthernet 0/0
ip address 192.168.30.1 255.255.255.0
no shutdown
exit
```

DHCP:

```
ip dhcp excluded-address 192.168.2.1 192.168.2.10
```

```
ip dhcp pool LAN_R1
network 192.168.2.0 255.255.255.0
default-router 192.168.2.1
exit
```

RIP:

```
router rip
version 2
no auto-summary
network 192.168.2.0
network 192.168.10.0
network 192.168.30.0
exit
```

```
end
write
```

7. Configure Router 2

Assume:

```
G0/0 → PC2
G0/1 → R0
Fa0/0 → R1
```

Enter:

```
enable
configure terminal
```

```
interface gigabitEthernet 0/0
ip address 192.168.3.1 255.255.255.0
no shutdown
exit
```

```
interface gigabitEthernet 0/1
```

```
ip address 192.168.20.2 255.255.255.0
no shutdown
exit
```

```
interface fastEthernet 0/0
ip address 192.168.30.2 255.255.255.0
no shutdown
exit
```

DHCP:

```
ip dhcp excluded-address 192.168.3.1 192.168.3.10
```

```
ip dhcp pool LAN_R2
network 192.168.3.0 255.255.255.0
default-router 192.168.3.1
exit
```

RIP:

```
router rip
version 2
no auto-summary
network 192.168.3.0
network 192.168.20.0
network 192.168.30.0
exit
```

```
end
write
```

8. Configure all PCs with DHCP

For **PC0**:

```
Desktop
→ IP Configuration
→ DHCP
```

For **PC1**:

```
Desktop
→ IP Configuration
→ DHCP
```

For **PC2**:

```
Desktop
→ IP Configuration
→ DHCP
```

You should get approximately:

PC0 → 192.168.1.x

PC1 → 192.168.2.x
PC2 → 192.168.3.x

9. Verify DHCP

On each router:

```
show ip dhcp binding
```

For example, R0 should show the IP assigned to PC0.

You can also check the PC:

PC0 → Desktop → Command Prompt

then:

```
ipconfig
```

10. Verify RIP

On R0:

```
show ip route
```

You should see something similar to:

```
C    192.168.1.0/24
C    192.168.10.0/24
C    192.168.20.0/24

R    192.168.2.0/24
R    192.168.3.0/24
R    192.168.30.0/24
```

The important thing is the **R**.

R = learned through RIP.

Your PDF uses `show ip route` specifically to verify RIP routes.

You can also check:

```
show ip protocols
```

11. Test RIP

From PC0:

```
ping <PC1's-IP>
```

and:

```
ping <PC2's-IP>
```

For example, if DHCP gave:

```
PC1 = 192.168.2.11
```

```
PC2 = 192.168.3.11
```

then:

```
ping 192.168.2.11
```

```
ping 192.168.3.11
```

You should get replies.

PART B — OSPF

Now you can create the **OSPF version of the same topology**.

You don't need to change the physical topology or IP addresses.

You only change the routing protocol.

Your PDF uses:

```
router ospf 1
network <network> <wildcard> area 0
```

for OSPF.

12. Remove RIP first

Do this on **each router**:

```
enable
configure terminal
no router rip
end
```

This removes the RIP configuration.

13. Router 0 — OSPF

```
enable
configure terminal
```

```
router ospf 1
network 192.168.1.0 0.0.0.255 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
exit
```

```
end
write
```

14. Router 1 — OSPF

```
enable
configure terminal
```

```
router ospf 1
network 192.168.2.0 0.0.0.255 area 0
network 192.168.10.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
exit
```

```
end
write
```

15. Router 2 — OSPF

```
enable
configure terminal
```

```
router ospf 1
network 192.168.3.0 0.0.0.255 area 0
network 192.168.20.0 0.0.0.255 area 0
network 192.168.30.0 0.0.0.255 area 0
exit
```

```
end
write
```

16. Verify OSPF neighbors

On R0:

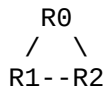
```
show ip ospf neighbor
```

R0 should have **two OSPF neighbors**:

R1

R2

That's because the topology is:



Every router has a direct OSPF relationship with the other two.

17. Verify the OSPF routing table

On R0:

```
show ip route
```

You should see routes marked:

0

For example:

```
C    192.168.1.0/24
C    192.168.10.0/24
C    192.168.20.0/24
```

```
O    192.168.2.0/24
O    192.168.3.0/24
O    192.168.30.0/24
```

0 means **OSPF**.

Your PDF's OSPF example likewise uses 0 in `show ip route` to identify an OSPF-learned route.

18. Test OSPF

From PC0:

```
ping <PC1-IP>
ping <PC2-IP>
```

For example:

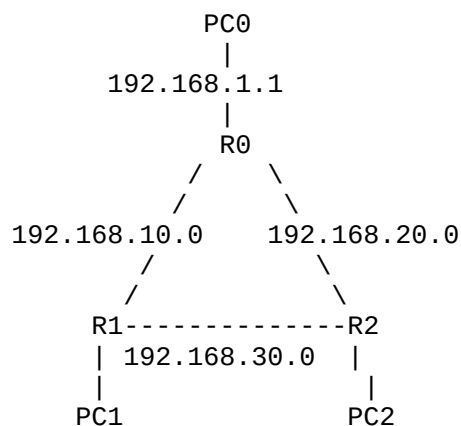
```
ping 192.168.2.11
```

```
ping 192.168.3.11
```

You should get successful replies.

Your complete mental picture

The final RIP network is:



Each router has:

1 LAN interface + 2 router-to-router interfaces = 3 interfaces.

And every router has its **own DHCP pool**:

```
R0 → 192.168.1.0/24 → PC0
R1 → 192.168.2.0/24 → PC1
R2 → 192.168.3.0/24 → PC2
```

Then:

RIP version

```
router rip
version 2
network ...
```

and routes appear as:

```
R
```

OSPF version

```
router ospf 1
network ... 0.0.0.255 area 0
```

and routes appear as:

```
O
```

One important practical point: if your 2901 does not show the exact FastEthernet0/0 interface I used above after inserting the module, don't force that command. Run `show ip interface brief` and use the **actual third interface name shown by your Packet Tracer router**. The IP addressing and RIP/OSPF commands stay the same.